

VLOOKUP in Excel

1. What is VLOOKUP?

VLOOKUP stands for Vertical Lookup. It searches for a value in the first column of a table and returns a value from another column in the same row. It solves the problem of finding related information, such as the salary of an employee or the price of a product, using an ID. A Data Analyst uses it to pull data from one table into another without searching by hand.

2. Syntax

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Argument	Required?	What it accepts
lookup_value	Required	The value to find, such as a number, text in quotes, or a cell reference
table_array	Required	The table to search (for example A2:D4). The lookup value must be in its first column.
col_index_num	Required	The number of the column to return, counted inside the table. The first column is 1.
range_lookup	Optional	FALSE for an exact match. TRUE for an approximate match (the default if you leave it out).

Use FALSE whenever you look up IDs, names, or codes. Use TRUE only for ranges, such as commission bands or grades, and only when the first column is sorted from smallest to largest.

3. Simple Example

Find the salary for employee ID 103 (exact match):

	A	B	C	D	E	F	G
1	ID	Name	Dept	Salary	ID		Salary
2	101	Rahim	Sales	30000	103		40000
3	102	Karim	HR	35000			
4	103	Hasan	IT	40000			

Formula in G2: =VLOOKUP(F2, A2:D4, 4, FALSE)

Excel finds 103 in the first column, A2:A4 (row 4). Then it returns the value from column 4 of the table, Salary, in that row. **Result: 40000**

4. Another Example

Find the commission rate for sales of 12000 (approximate match):

	A	B	C	D	E
1	Sales From	Rate		Sales	Rate
2	0	0%		12000	5%
3	5000	2%			
4	10000	5%			
5	20000	8%			

Formula in E2: =VLOOKUP(D2, A2:B5, 2, TRUE)

12000 is not in the list. With TRUE, Excel finds the largest value that is not more than 12000, which is 10000, and returns its rate. **Result: 5%**

5. When to Use It

- Find an employee's salary or department using an employee ID
- Retrieve a product name or price using a product ID
- Match student IDs to marks or grades
- Apply commission rates, tax slabs, or grade bands (approximate match)
- Bring customer details into an orders table

6. Important Things to Remember

- The lookup value must be in the first column of table_array. VLOOKUP only returns columns to the right of it.
- col_index_num counts columns inside the table, not sheet column letters. In B2:E10, column 1 is B.
- If range_lookup is left out, Excel uses TRUE (approximate), which can give wrong results for IDs. Always type FALSE for exact matches.
- With TRUE, the first column must be sorted smallest to largest. Excel returns the largest value that is not more than the lookup value.
- If the value is not found, the result is #N/A. If col_index_num is larger than the table, the result is #REF!.
- If the first column has duplicates, VLOOKUP returns the first match. Matching is not case-sensitive.

7. Common Mistakes

Mistake: Leaving out FALSE, so Excel uses approximate matching and returns the wrong row. **Fix:** Always end an exact lookup with , FALSE).

Mistake: Copying the formula down without locking the table, so the range moves and gives #N/A. **Fix:** Use \$ signs: \$A\$2:\$D\$10.

Mistake: Having the lookup value outside the first column of the table. **Fix:** Start the table at the column that holds the lookup value, or use XLOOKUP or INDEX + MATCH.

Mistake: Getting #N/A for an ID that looks correct, because one is stored as text or has extra spaces. **Fix:** Make both sides the same type and clean the data with TRIM.

8. Data Analyst Tip

Put the lookup value in a cell, lock the table with \$, and wrap the formula in IFERROR so missing IDs show a clear message:

=IFERROR(VLOOKUP(F2, \$A\$2:\$D\$4, 4, FALSE), "Not Found"). In Excel 2021 or Microsoft 365, XLOOKUP is easier for new work because it needs no column number and can look to the left.